HI-LO HIGHWAY

iGaming Card Prediction Game · Original Concept

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1. Executive Summary

Format: single-screen HTML5 game · mobile-first · card prediction with cash-out
RTP: 98.0%, exact for any strategy and any cash-out point
Genre: Hi-Lo / escalation, where the player picks the risk of every step
Board: 4 piles, each growing up to 5 cards (16 guesses max, 20-card full board)
Max win: ×10,000; any guess that would exceed it is disabled
Signature feature: the board persists after cash-out, and the player can buy back into it at a price that scales with its progress

2. Core Concept & Theme

2.1 Concept

Four cards are dealt face up, one per pile. On each turn the player picks a pile and calls Higher or Lower. A new card lands on that pile: right call, the multiplier climbs; wrong call or same rank, the round is lost. The player can cash out at any time. Picking the pile is the decision that matters: an Ace or a King is a near-safe step (×1.08), a 7 is a coin flip (×2.17), a Queen called Higher is a 1-in-13 shot (×13). The full board (20 cards) triggers an automatic cash-out.

After a cash-out, the board stays on the table. The player can start a fresh board at the base stake, or **resume this board**: the stake is multiplied by the multiplier reached, and the run continues from the same cards. The further the board went, the more it costs to step back in, and the bigger the remaining upside. The price is exactly fair, so RTP stays at 98%. What sells it is the pull of a "hot board", even though every draw is independent.

2.2 Visual Direction

Kaching Originals house style, shared with Ricochet and Candy Monster: dark blue table (#0b1622), panels in #172533, green action button (#00e701), Inter font. Flat, high-contrast cards readable on a phone, with suits as decoration only. The four piles fan out vertically as they grow, so the board itself shows how far the run went.

2.3 Competitive Positioning

	Stake Hilo (Stake Originals)	Hi-Lo Highway
RTP	99%	98%
Cards in play	1 current card	4 persistent piles
Same rank	wins with "or same"	loses
Skip	yes	no; the pile choice replaces it
Run goal	open-ended	full board, 16 steps
Re-entry	no	buy back the board at a fair price

3. Game Mechanics

3.1 Round Flow

- A board of 4 face-up cards is already on the table. The player sets the bet and presses **Bet**: the stake is deducted.
- Under each pile, two buttons show the call (▲ Higher / ▼ Lower), the multiplier the player would reach and the chance of winning.
- The player taps one button. A new card is dealt onto that pile. Right call: the multiplier and the potential win update. Wrong call or same rank: the round is lost and the stake is gone.
- After the first right call, **Cash out** pays bet × multiplier. Each pile holds 5 cards; when all four piles are full, the win is cashed out automatically.
- After a cash-out: **New board** (base stake, fresh deal) or **Resume this board** (fair buy-back price, same cards, multiplier carried over). After a loss, the next round deals a new board.

3.2 Rules

- Ace is low (1), King is high (13). Suits are cosmetic.
- Infinite shoe: every draw is uniform over the 13 ranks and independent of the cards on the board.
- Same rank loses. "Higher" on a King and "Lower" on an Ace can never win, so these buttons are disabled.
- A guess whose multiplier would exceed ×10,000 is disabled and shows MAX. If no guess remains, the win is cashed out automatically.

3.3 Board Buy-Back

After a cash-out at multiplier m on base stake b, the player can resume the same board. The buy-back price is **b × m ÷ 0.98**. The run continues with the multiplier still at m (payouts stay quoted on the base stake), the same top cards and the same remaining space on each pile. A full board cannot be resumed.

The price equals the expected value of the position divided by the RTP, so a resumed run returns exactly 98% of what it costs, like a fresh one. The buy-back changes the size of the bet and the shape of the board the player keeps, never the edge.

3.4 Interface

- Header: current multiplier (dimmed before the first right call, green on a win, red LOST on a loss) and a one-line status: potential win, cards found, reason for a loss.
- Piles fan downward; the newest card is in front. A dashed outline shows where the next card will land.
- Each guess button shows the resulting multiplier and the probability, so the player always sees the risk he is buying.
- Cash-out button shows the amount; the buy-back button shows the price and the multiplier carried over.
- History of the last 16 rounds, session stats (rounds, cash-outs, best win), odds table per card under the game.
- Shortcuts: Space to bet, Enter to cash out. French and English.

4. Mathematical Model — Constant-RTP Ladder

4.1 Odds per Card

Chance of winning each call and the step factor (1 ÷ chance) applied to the multiplier:

Top card	Higher	Lower
A	92.3% · ×1.08	—
2	84.6% · ×1.18	7.7% · ×13.00
3	76.9% · ×1.30	15.4% · ×6.50
4	69.2% · ×1.44	23.1% · ×4.33
5	61.5% · ×1.63	30.8% · ×3.25
6	53.8% · ×1.86	38.5% · ×2.60
7	46.2% · ×2.17	46.2% · ×2.17
8	38.5% · ×2.60	53.8% · ×1.86
9	30.8% · ×3.25	61.5% · ×1.63
10	23.1% · ×4.33	69.2% · ×1.44
J	15.4% · ×6.50	76.9% · ×1.30
Q	7.7% · ×13.00	84.6% · ×1.18
K	—	92.3% · ×1.08

4.2 Multiplier

After n right calls with winning chances $p_1 \dots p_n$, the multiplier is **$m = 0.98 \div (p_1 \times p_2 \times \dots \times p_n)$**. Each step multiplies the expected value by $p \times (1 \div p) = 1$, so the expected payout of any cash-out is 0.98 × the stake, whatever the piles chosen, the calls made or the moment of cash-out. The RTP is therefore exactly 98% for every strategy, including the buy-back (section 3.3).

Example run on a 1 € stake:

Call	Chance	Step factor	Multiplier
Higher on Ace	92.3%	×1.08	×1.06
Lower on 7	46.2%	×2.17	×2.30
Higher on Queen	7.7%	×13.00	×29.90

Cashing out after these three calls pays 29.90 €. The chance of getting there was 3.28%.

4.3 Cap and Full Board

Guesses that would push the multiplier past ×10,000 are not offered, so the cap never cuts a payout and never lowers the RTP. The full board (16 right calls) and the "no guess left" case both cash out at the multiplier reached.

4.4 Simulation

Five strategies simulated on the same rules as the prototype, 19 million rounds in total (95% confidence intervals):

Strategy	Rounds	RTP
Prudent: safest call, cash out after 6	4,000,000	98.08% ± 0.22
Risky: longest shot, cash out after 2	4,000,000	97.94% ± 0.77

Strategy	Rounds	RTP
Random call, cash out after 4	4,000,000	97.18% ± 0.93
Full board: safest call every time	4,000,000	97.97% ± 2.65
Random, 3 calls, then 2 buy-backs	3,000,000	97.41% ± 1.92

Every result is consistent with 98%. Wide intervals come from rare very large multipliers, which is also why the RTP is proven analytically (4.2) rather than only measured.

5. Audio Design

- Card deal: short filtered noise flick.
- Right call: two-note triangle chime that rises in pitch with the length of the run.
- Wrong call: falling sawtooth tone.
- Cash-out: four-note ascending arpeggio.
- All sounds synthesised with the Web Audio API; Sound ON / OFF toggle.

6. Technical Specifications

- One HTML file: DOM cards and CSS animations (deal, win glow, loss shake), vanilla JavaScript, no image assets.
- RNG: crypto.getRandomValues; each card is drawn at the moment of the call and shown immediately.
- Responsive layout from 320 px phones to desktop; card size computed from the board width.
- French and English interface; play-money balance shared with the other portfolio games.

7. Commercial Roadmap

- RGS integration with server-side outcome generation and a certified RNG.
- Math certification (GLI-19 / BMM) of the ladder and of the buy-back pricing.
- RTP variants by changing the 0.98 factor (for example 97% or 96%).
- Finite-shoe variant (6 decks, cards removed as they are dealt), which turns the board into a light card-counting puzzle and needs its own pricing.
- Shared table: boards cashed out by one player offered to others at the buy-back price.